STAT 330-001

40142034 58

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Test: 641391 Form: A Wadsworth, Spencer Gordon

Entry: 06:59pm 10/21/2026 Checked out:

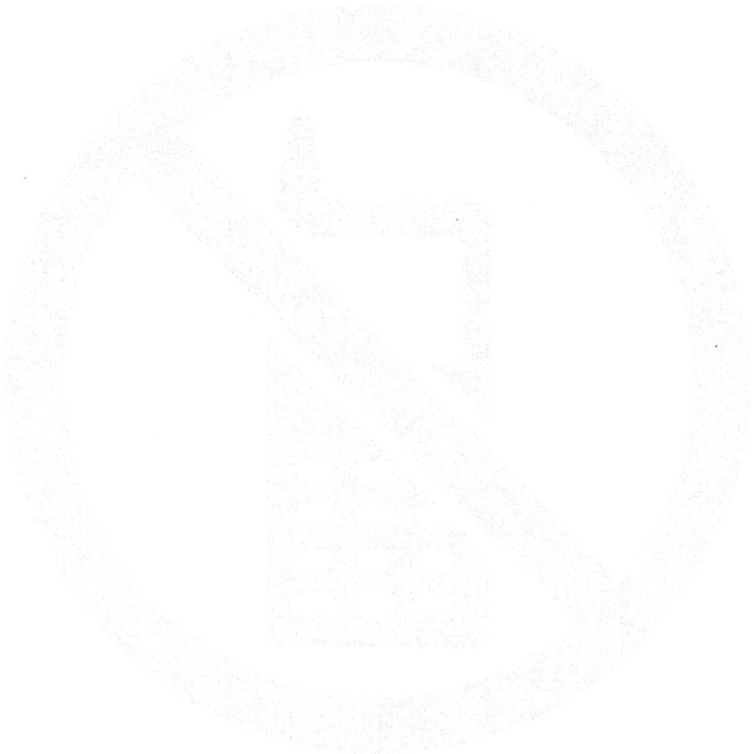
Employee: Ryan A.

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE



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058

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

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- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
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PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

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(A) always increases or remains unchanged.

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Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

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(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

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(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

LASSO and ridge regression are sometimes referred to as shrinkage methods because they shrink the standard error of the coefficients, allowing for better prediction. They do this by shrinking the fitted coefficients themselves towards zero (or, for LASSO, sometimes eliminating them), which deliberately introduces bias but creates a narrower interval for prediction. LASSO seeks to minimize $|\beta_i|$, and ridge seeks to minimize β_i^2

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right) = \frac{5.76}{4}\left(\frac{.25}{.75}\right)$$

(Circle one answer) Which conclusion is most appropriate?

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(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

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(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

β is some fixed, unknown, population parameter (or vector of parameters) of a linear coefficient, while $\hat{\beta}$ is a fitted estimate of β that we can use to draw inference

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

Conceptually, the sum of squared errors measures how far off our model is from the actual values. Least squares asks, "for what values of $\hat{\beta}$ is the sum of squared errors the smallest?" i.e. for which $\hat{\beta}$ does the model get as close as possible to the actual data?

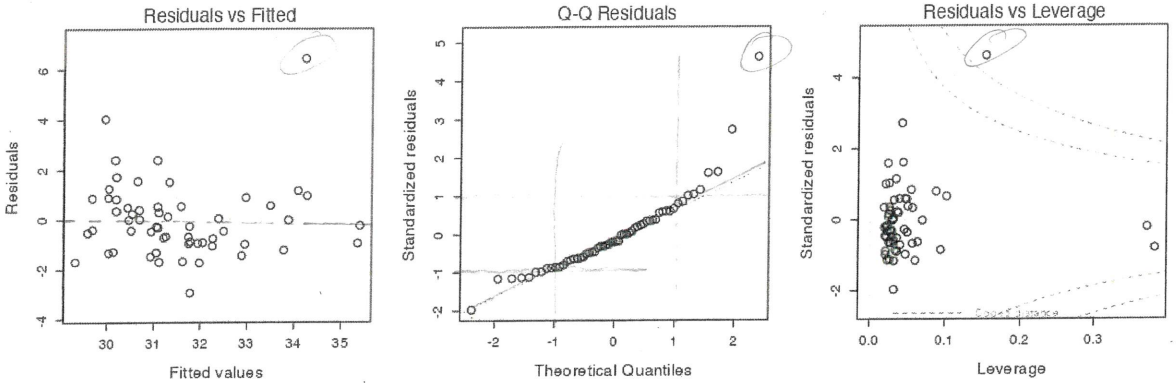
you have no idea how funny this was in the quiet testing center

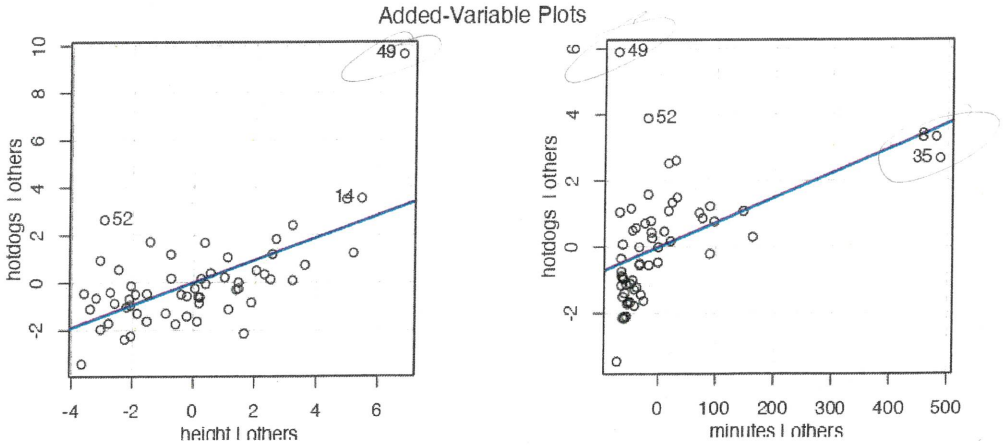
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

$\beta_1 = .473$

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

$\hat{\beta}_1 = .473$. Holding all other predictors constant, a one-inch increase in height is associated with an increase in eating capacity of .473 hot dogs in a ten-minute sitting

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Not overly well. From the AV plots, height seems to be linear, but minutes does not, and there is a slight downward trend in the residuals vs. fitted plot

(B) There is independence between each observation

Plausible, but with some problems if we want to generalize. Perhaps individuals of a certain height are more likely to become eaters. Additionally, it was not a random sample, because the eaters volunteered. For eaters specifically, however, perhaps it is fine.

(C) The errors are normally distributed

This mostly looks fine, but there are some serious problems towards the ends of the Q-Q plot - the errors appear to be slightly right skewed with errors compressed on the left and stretched on the right. The Shapiro-Wilk test is less informative, but also a very small p-value.

(D) The errors have constant variance (homoscedasticity)

We have problems here. There is a clear funnel shape on the fitted vs. residual plot, and although the Breusch-Pagan test p-value is not significant, it's possible that this is due to the outliers.

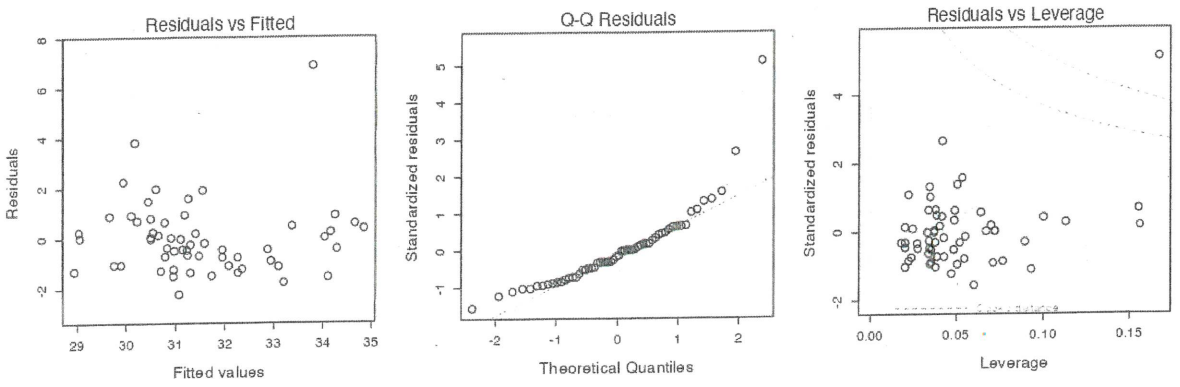
(E) There are no unduly influential points

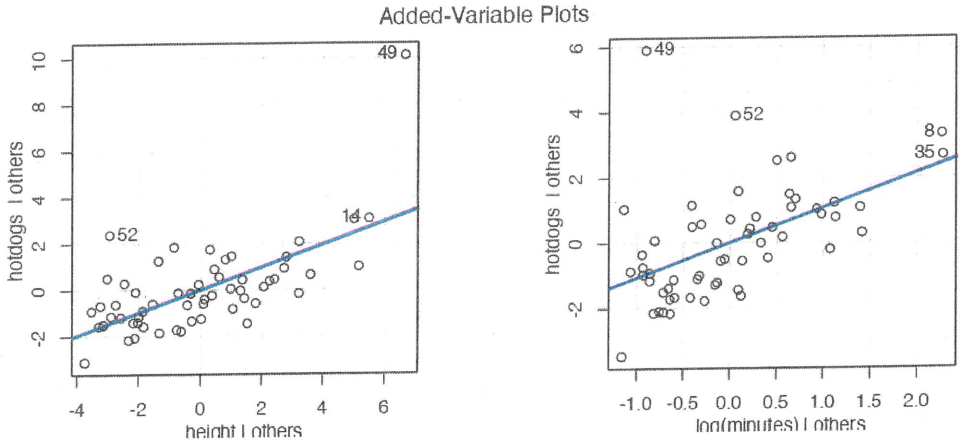
It appears that there are multiple unduly influential points. As shown in the residuals-vs.-leverage plot, there is one observation with a Cook's distance greater than 1, and two others that approach .5, making these points potentially unduly influential.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

Not necessarily. How do we define "superior"? Perhaps model (1) is more likely to satisfy the equal variance assumption than model (2). But we should consider other assumptions, interpretability, and how well the models actually fit before throwing around words like "clearly superior".

Mr. Big Shot Statistician

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Advantages: Model (2) seems to satisfy the linearity assumption better, particularly for minutes (the log vs. the raw). It seems marginally more normal (except the $r_i = 5$ outlier). From the residuals vs. leverage plot, its Cook's distances seem smaller than model (1), and its CV RMSE is indeed smaller.

Disadvantages: Model (2) loses interpretability with the log transformation. The equal variances problem actually appears slightly worse than in model (1), from both the plot and the Breusch-Pagan test. Finally, its standard error for minutes is much higher, which means less reliable predictions.

Question 16: State the estimated value of β_2 and provide an interpretation in context

$\hat{\beta}_2 = 1.067$. Holding all other predictors constant, a 1% increase in minutes per week spent training for an eating contest is associated with an increase in mean consumption of about .011 hot dogs in a ten-minute sitting.

$1.067 \ln\left(1 + \frac{1}{100}\right) = .0106$

$1.067(.01) \approx .01067$

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\hat{y}_f = -6.05589 + .48834(68) + 1.067\log(79)$

$= -6.05589 + 33.20712 + 4.661723$

$= 31.813$

f for friend

model (2) predicts that my friend can consume 31.8 hot dogs in 10 minutes

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

$58 - 2 - 1 = 55$

(A) 55

(B) 57

(C) 56

(D) 48

this is prediction

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$31.813 \pm 2.004(1.485)$ (28.837, 34.789)

± 2.976

95% prediction interval : (28.837, 34.789)

Part III - Application

Questions 20-24 concern the following dataset:

In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

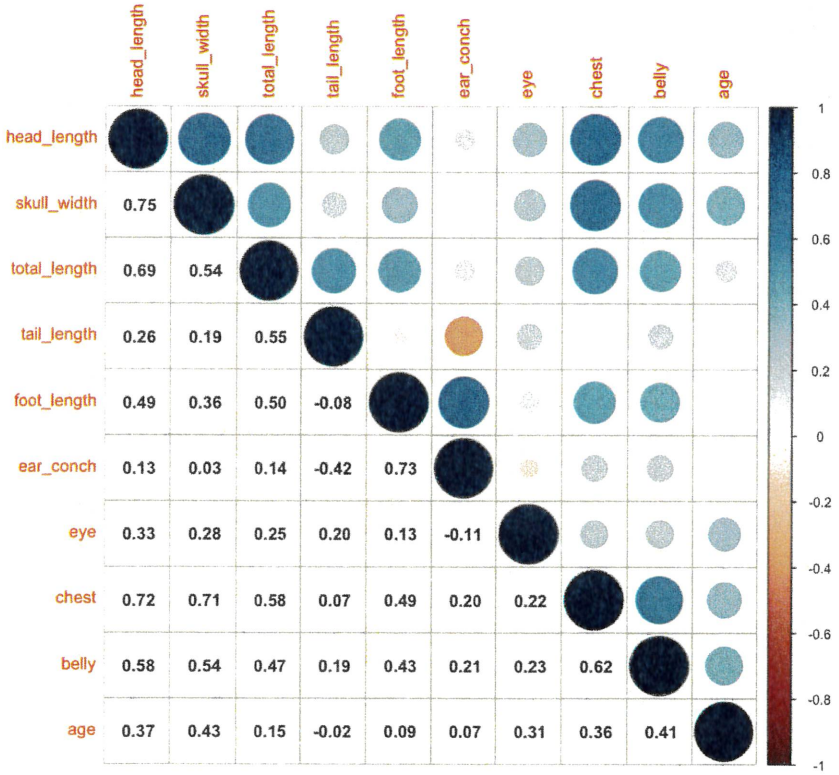
Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

.16559 .10599

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

.16559 ± 2.01(.10599)

.16559 ± 2.01(.10599)

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that for this model with all predictors the interval (-.023, .443) captures the true coefficient of belly width, β_{belly}.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

The overall F-test has as its alternate hypothesis that at least one slope is not 0; it tests the model as a whole. Although for any one predictor, there might not be a linear relationship with age, some combination of predictors might have one, fueling the low p-value.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The best subsets algorithm considers every possible multilinear model with the given predictors, and calculates the AIC (a metric for model fit that prioritizes prediction and only mildly penalizes lots of variables) for each one. The model with the lowest AIC gets chosen as the best one under this criterion.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report 6

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE 7

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE 5

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data 3

A Initialize vectors for each model that you will use to store their RMSEs 1

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets 2

B Use the models fitted to the training data to get predictions for the test data 4

END OF EXAM

STAT 330-001

40142034 56

Morehead, Ethan Timothy

279452696 ethan0 385-427-4418

Test: 641236 Form: A Wadsworth, Spencer Gordon

Entry: 06:17pm 10/21/2026 Checked out:

Employee: Jemma C.

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Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Lasso and ridge are sometimes referred to as "shrinkage" methods because they shrink the $\hat{\beta}$ of less significant terms by add to the other parts of the model.

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$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

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$\hat{\beta} = \frac{Y-\epsilon}{X}$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

$\hat{\beta}$ is the estimated value of β, while β is the true value. (β is found from the population.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

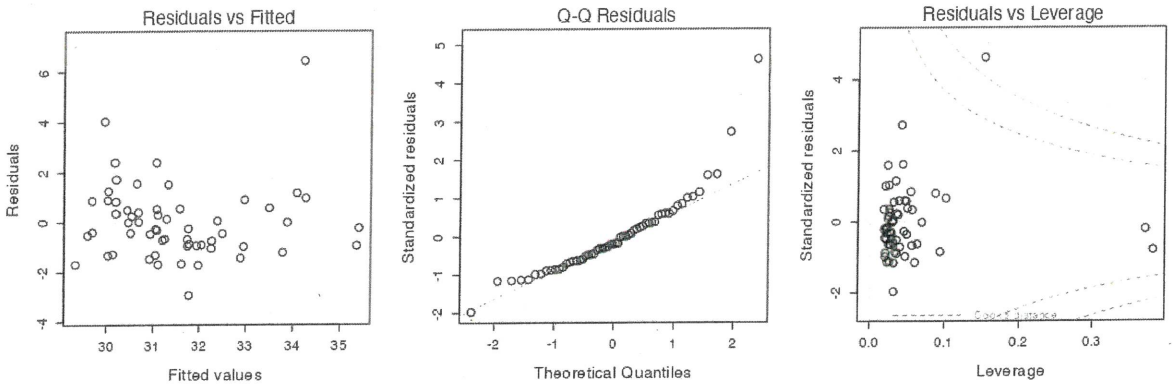
By comparing the least squares we can estimate the regression coefficients by minimizing the least squares.

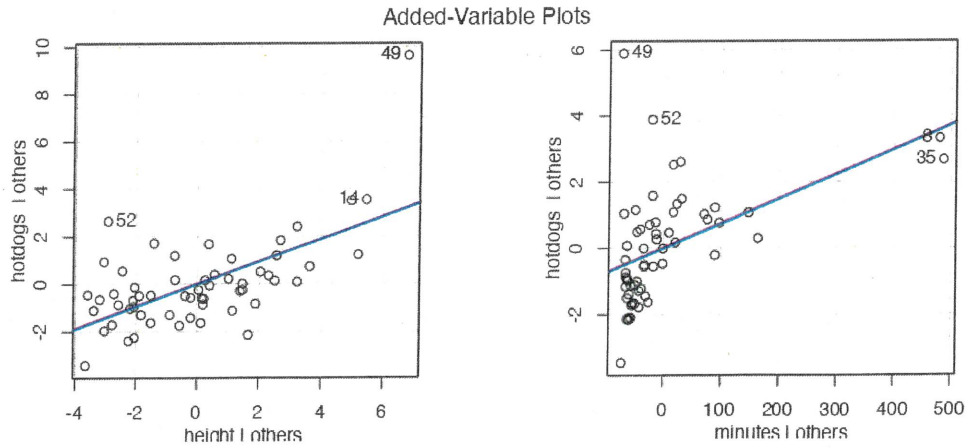
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	Pr($>\|t\|$)
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

The estimated value of β_1 is 0.472892. If a group of people were selected and all other variables were constent we could expect the average number of hotdogs to increase by 0.472892 for each inch.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

It doesn't have a very linear relationship with minutes. If more data was pulled than it might even out but right now there are two groups of responses that split the group.

(B) There is independence between each observation

There apears to be independence between the observations

(C) The errors are normally distributed

the errors are not normally distributed

(D) The errors have constant variance (homoscedasticity)

Yes the errors have constant variance.

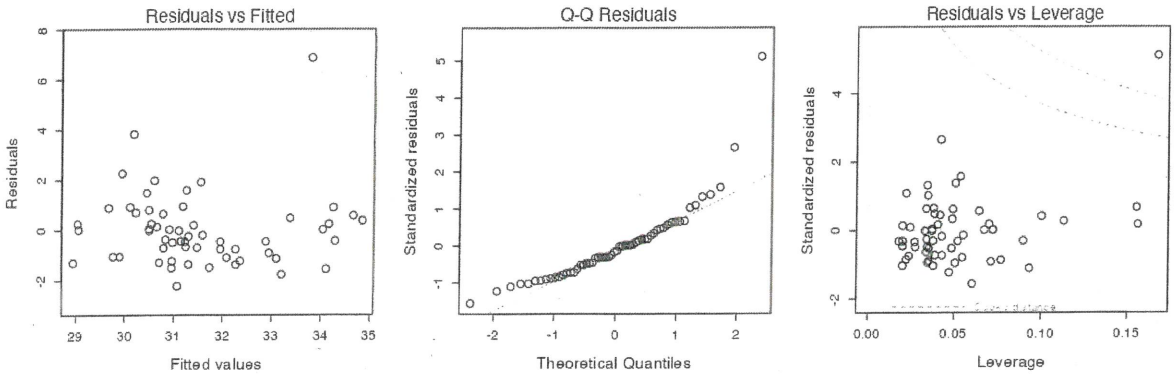
(E) There are no unduly influential points

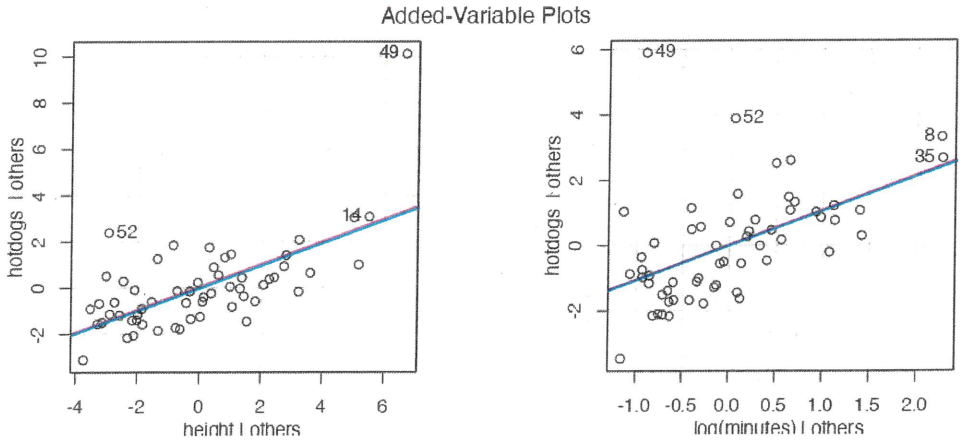
This is false. There are multiple points that are rather far away from the others.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would show him the graphs and explain the scewiness that is being implements into model(1) & the show model 2.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Model (2) better fits our linear assumptions we make.
Model (1) does have a lower standard error.
Model (2) also has a higher Breusch-Pagan p-value.

Question 16: State the estimated value of β_2 and provide an interpretation in context

The estimated value of β_2 is 0.48834

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\hat{Y} = -6.05589 + 68 \cdot 0.48834 + \log(7) \cdot 1.06702$

$\hat{Y} = 31.813$

31.813 hotdogs

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

$58 - 2 - 1 = 55$

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$\hat{\beta}_i \pm t^* SE(\beta_i)$

31.813 ± 1.485

$= (30.328, 33.298)$

Part III - Application

Questions 20-24 concern the following dataset:

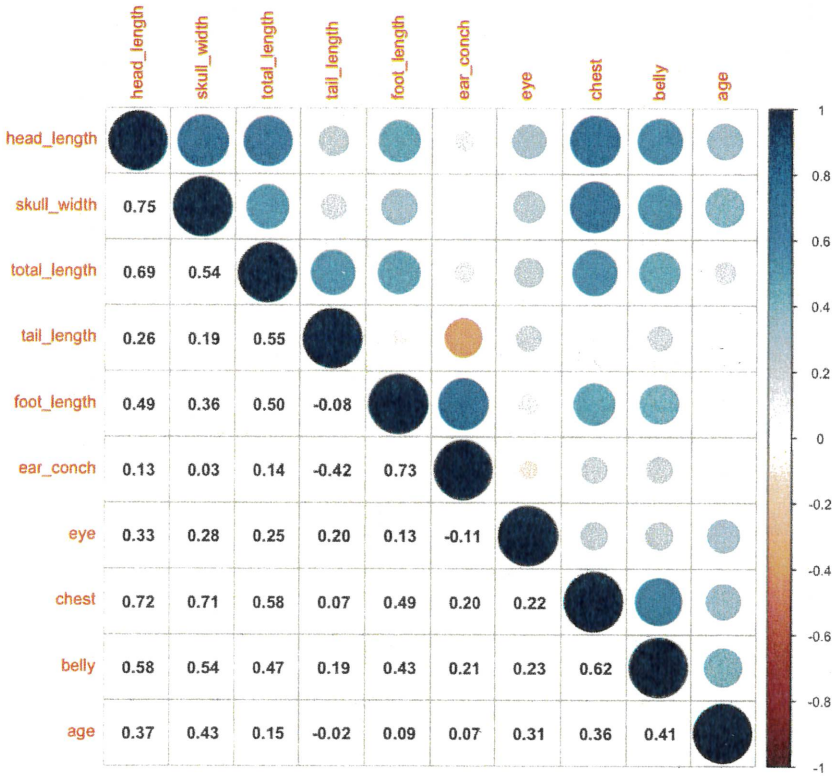
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20: 0.16559

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$\hat{\beta}_i \pm 2.01\ SE(\hat{\beta}_i)$

$0.16559 \pm 2.01 * 0.10599$

$0.16559 \pm 2.01 * 0.10599$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

Holding all else constant (and not having an influence) the belly girth can predict the age changing it by between -0.023 & 0.443 years for every cm of the belly girth with 95% confidence.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

Each value by itself is not enough to have a significant effect but when combined a pattern emerges.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The "best subsets algorithm will start by taking the factor that is most significant & then searching for another that is also significant & the repeat.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

D Average RMSE across all test-training splits and report

C Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

C Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

C Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 55

Hunsaker, Colin

390856759 gacolin 678-895-5027

Test: 641175 Form: A Wadsworth, Spencer Gordon

Entry: 06:05pm 10/21/2026 Checked out:

Employee: Kaitlyn P.

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

Colin Hunsaker

NS

40142034

055

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Evidence of this on page 6

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Variance is being shrunk because Bias is being added.

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$\frac{2.4^2}{4} = 1.44\left(\frac{.25}{.75}\right) = .48$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

β is the true coefficient predictor effect for a given response variable. $\hat{\beta}$ is our best estimate of that true predictor given our data + model that we make.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

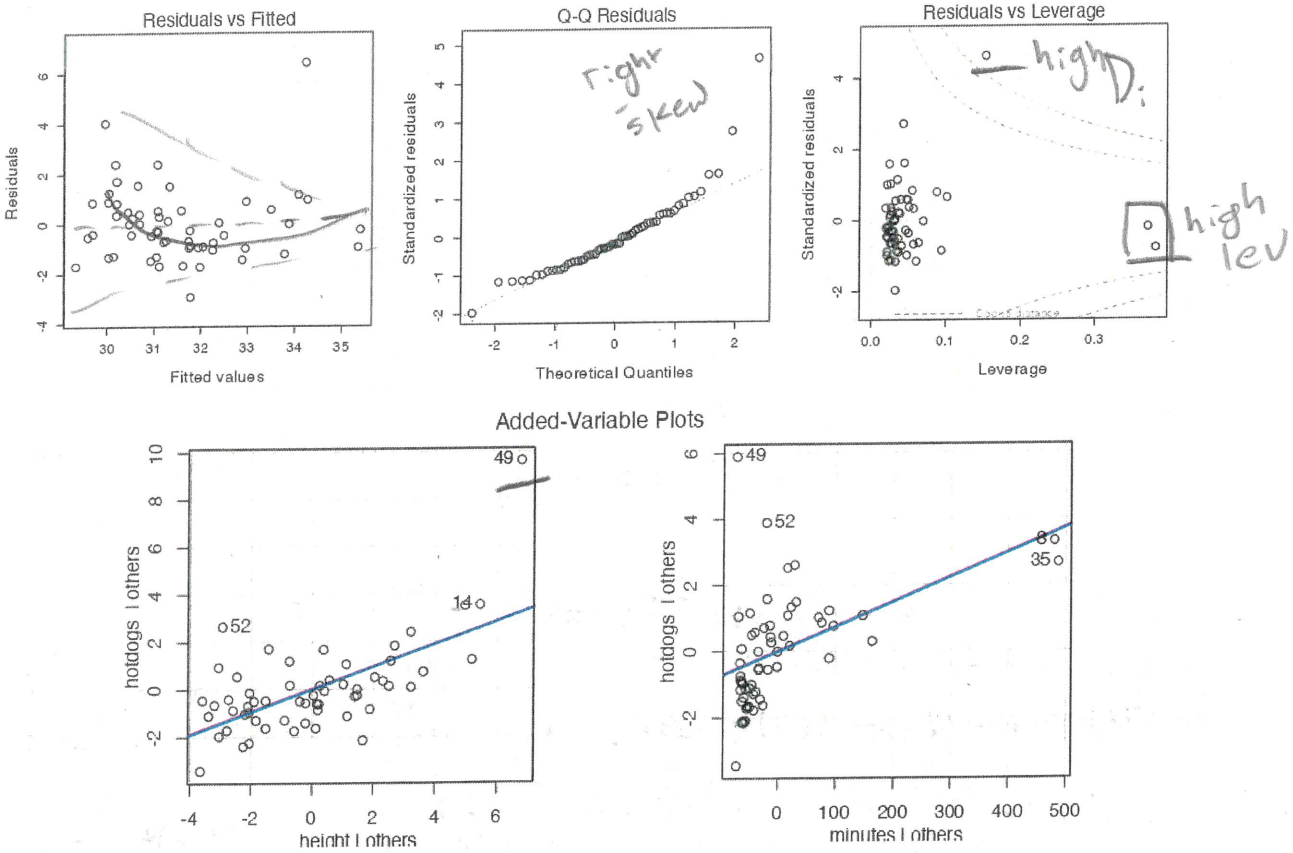
LS estimates regression coefficients by taking all of the data and formulating the coefficients (finding the $\tilde{\beta}$s) that minizes the total least squares error for all observations.

PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$



Coefficient	Estimate	Std. Error	t value	Pr($>\|t\|$)
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

estimated β_1: $\hat{\beta}_1 = .472892$

Holding minutes per week training constant, on average a 1 inch increase in height resulted in an increase of .472892 hot dogs consumed in the 10 min sitting.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

$.007417 - 1.995 \cdot .001871 = .003684$

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$ → that's not how p-values work

(D) $\beta_1 = 0.4729$ → Beta

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

In the residual v fitted plot, the data is relatively evenly disperced around 0. It's not the best, but fine. The added variable Plot For hd and height looks incredibly linear, while hds and minutes looks fine. Not super worried. Linearity seems okay.

(B) There is independence between each observation

Because the data comes from seperate individuals, and we assume to contest is Fair, we can assume independence between each observation.

(C) The errors are normally distributed

This one is not too good. The QQ plot has a U shape, meaning the data is likely right-skewed. This concern is also backed up w/ the significant Shapiro-Wilk test p-value, we will proceed w/ caution.

(D) The errors have constant variance (homoscedasticity)

This one isn't the best either. There is a clear funnel shape to the residuals vs fited values plot. The Breusch-Pegan test p-value is close to significant on $\alpha = .05$ → we will proceed w/ caution

(E) There are no unduly influential points

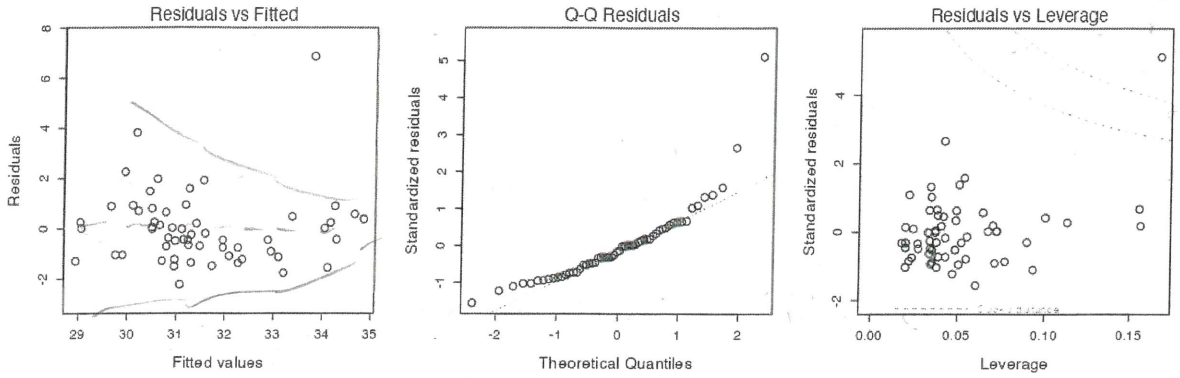
There are a few potentially influential points. Two points have high leverage, while one has a particularly large Cook's Distance. We will note them and review them before preceeding w/ caution

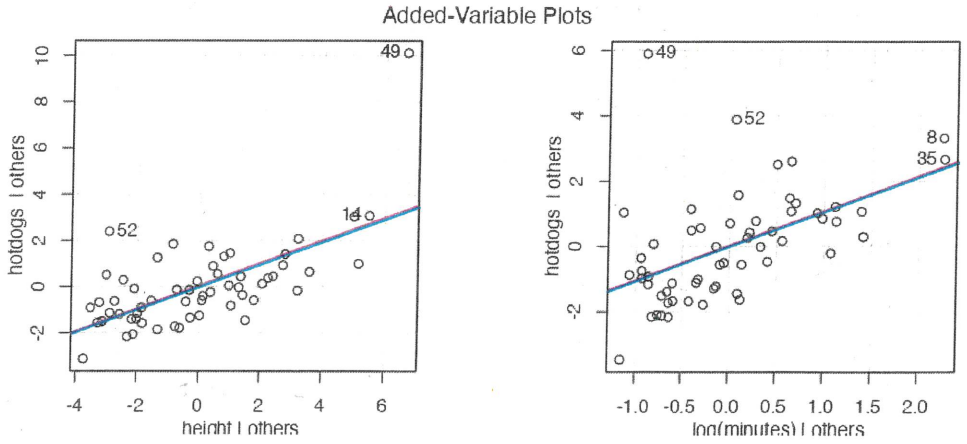
Model 2

a level-log transformation!

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
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Breusch-Pagan p-value	0.028
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CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would tell the reseacher to get some rest. A Breusch-Pagan test is not an end all be all in determining which model is superior. The change was not huge either. IF we look at the plots, we can see legitimate improvement in model 2 For leverage, and linearity for hotdogs vs log(minutes). Statistics is too-vibes based to draw a conclusion from one small change in a p-value for a B-P test.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Advantages: Model (2) appears slightly more normal (larger p-value for S-W, less lines leaving the Quantile line in the Q-Q plot), while limiting the leverage of the high-leverage observations mentioned before, while greatly improving the linearity assumption (less pattern around the residual v Fitted, added var for hotdogs v log(mins) is much more linear).

Disadvantages: Less equal variance (lower p-value for the Breusch-Pagan test, a stronger funnel shape in the residuals v fitted than in model 1) and harder to interpret log(mins)

Question 16: State the estimated value of β_2 and provide an interpretation in context

Holding all other predictors constant, a C% increase in minutes, gives a log(1+C/100) increase in the mean number of hot dogs consumed.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

intercept $\hat{\beta}_1$ $\hat{\beta}_2$

$-6.05589 + .48834 \cdot 68 + 1.06702 \cdot 4.369 = 31.81304$ hot dogs

I would predict that my friend could consume around 31.81304 hot dogs. If I were given additional information, I could form a prediction interval, which would be wider than a confidence interval.

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

it is

$n - p - 1$

$58 - 2 - 1$

55

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

I was just talking about this!

$31.81304 \pm 1.485 \cdot 2.004$

31.81304 ± 2.97594

$(28.8371, 34.78898) \rightarrow$ 95% prediction interval

95% of the time this prediction interval will include the number of hot dogs my friend consumes

Part III - Application

Questions 20-24 concern the following dataset:

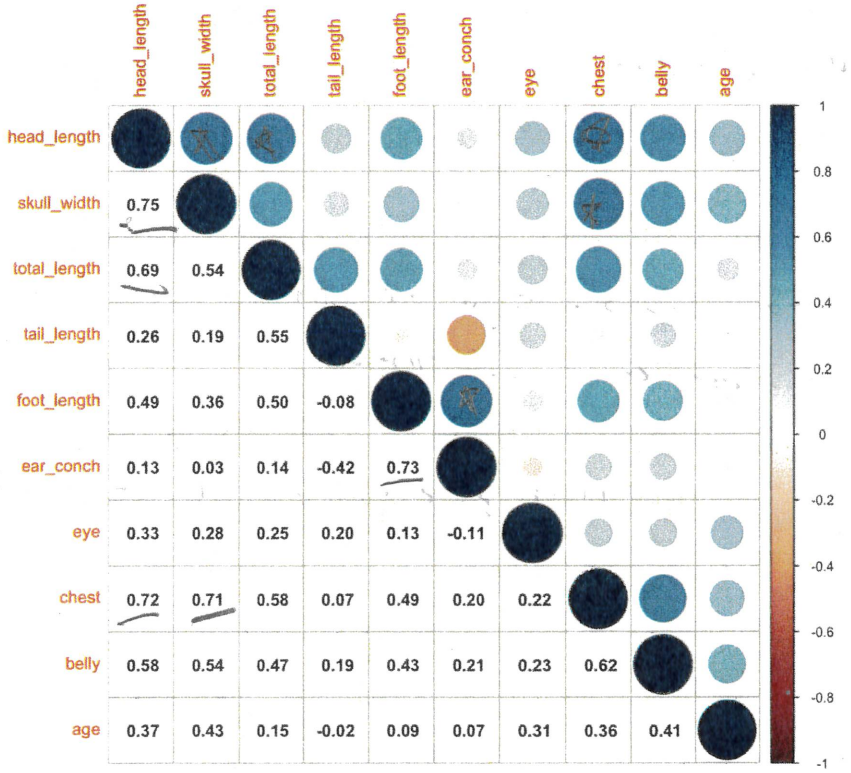
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$.16559 \pm 2.01 \cdot .10599$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

If we repeat this entire sampling and analysis process many many times, 95% of the time the β_{belly} coefficient is between -.023 and .443 OR we are 95% confident the true coefficient for belly width is between -.023 and .443

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

The F-test only test if any predictor is significant. we don't know which, if any variables are actually significant. Many of the variables are close to significant on the $\alpha = .05$ level, drawing the F-test to show a significant p-value.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

"Best subsets" is an algorithm that reviews every single possible of the $2^9 = 512$ models, then selects the model with the lowest AIC metric. (A metric used for large, predictive models, that does not punish the # of predictors used)

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

adjusted R^2 has a lower penalty than AIC. We don't know the significance cut off that would result in increasing R^2. It is possible that the p-value of .04287 could be not strong enough of a predictor to increase adjusted - R^2

Holy Cow

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

B Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

B Compute each model's AIC and BIC

D Fit each model using the testing data → test against this data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

C Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 54

Bake, Bryce Dean

868423142 bbake25 208-590-8415

Test: 641174 Form: A Wadsworth, Spencer Gordon

Entry: 06:04pm 10/21/2026 Checked out:

Employee: Christian S.

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly.
Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

40142034
054

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Lasso shrinks the model by shrinking some coefficients to 0. This helps eliminate noise or weak predicto coefficients.

Ridge also shrinks coefficients but not to 0. That way all of them are small but still have some influence.

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^T X)^{-1} X^T y$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

$\hat{\beta}$ is the coefficient prediction from the sample set we used.
β is the coefficient of the theoretical population parameter we don't know.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

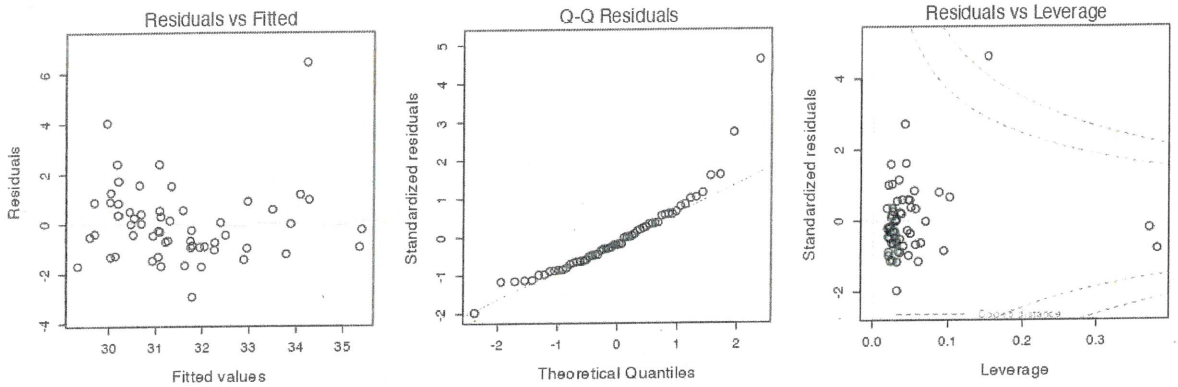
The least squared is helpful because it takes away negative values by squaring and also punishes larger numbers.

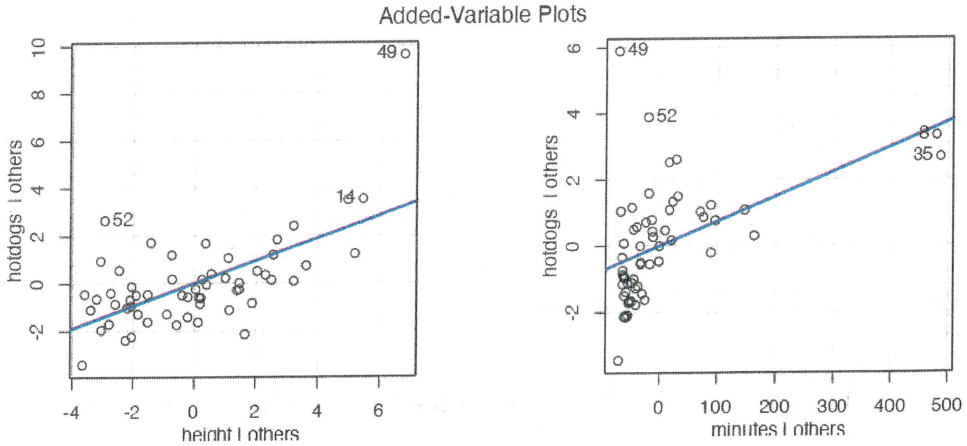
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	Pr($>\|t\|$)
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

Estimated $\beta_1 = 0.472892$

Holding minutes constant, a 1 inch increase in height results in a 0.472892 increase in the amount of hotdogs eaten.

Question 12: **(Circle all that apply)** Which of the following are true? $0.007417 \pm (3.963)(0.001871)$

$0.00741 \quad 0.00741 \quad 4773$

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

The line seems to have disappeared but the R vs. f plot seems to have a linear line. The added variable plots also seem linear in the positive direction, however minutes is definitely worth looking into.

Yes, linear.

(B) There is independence between each observation

Yes, each person/contestant is independent of the other meaning their results indipendent as well.

(C) The errors are normally distributed

Looking at the Q-Q plot it seems to vary from the line a lot at the top. It seems to be right skewed and we can't conclude that the errors are normally distributed.

(D) The errors have constant variance (homoscedasticity)

Using the Residual Plots the observations seem to make a column shape and are spread out suggesting heteroscedasticity or non constant variance.

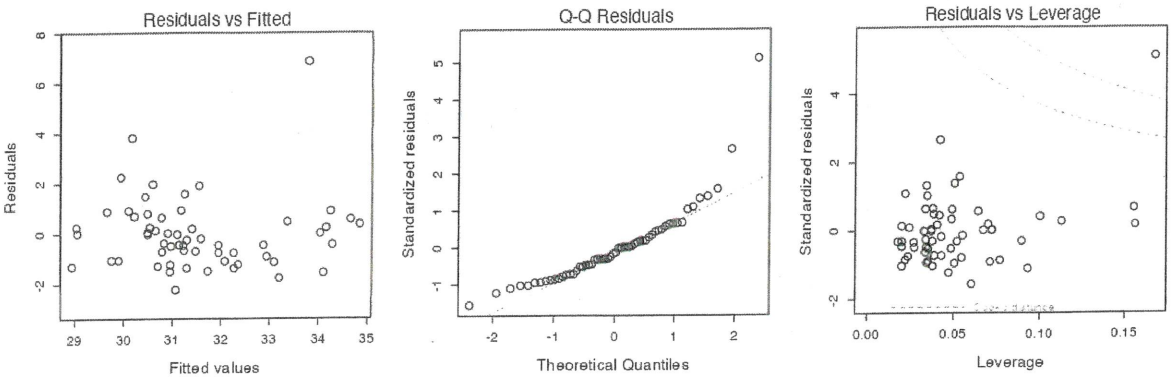
(E) There are no unduly influential points

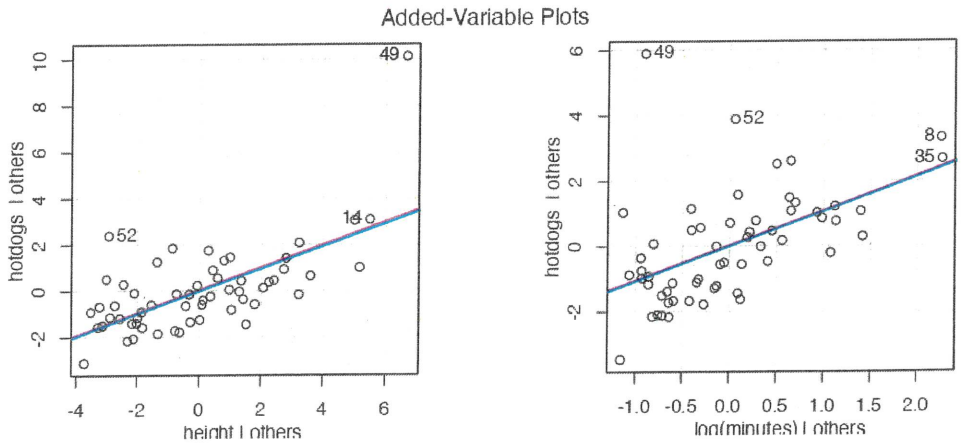
Since there are no cook's test included in the other summaries for only specific observations it seems there aren't unduly influencial points. 49 is very high but since it is not far right on the X axis it is okay.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would respond that is true, however, The Cross Validation error and RMSE on model 2 is smaller making that one more attractive. The p-values for our intercept and predictors are also smaller showing model 2 might be better to use.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Advantages - Smaller CV RMSE of 1.66702, smaller p-values for predictors, Added Variable plots are more linear.

Disadvantages - Breusch-Pagan p-values is .028, The Residuals vs leverage plot is more spread out suggesting unconstant variation so results are less certain than model 1.

Question 16: State the estimated value of β_2 and provide an interpretation in context

Estimated $\beta_2 = \log(1.06702)$ $e^{1.06702} = 2.9$

Holding height constant for every additional 1 minute there is an increase of 2.9 hotdogs eaten by a contestant on average.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$-6.05589 + 0.488334(68) + 1.06702(4.369) = 31.8$ Hotdogs in 10 minutes.

$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 H + \hat{\beta}_2 \log(M)$

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55 $n-p-1$

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

Estimate $\pm$ t^* SE

$0.204 \pm 2.004(1.485)$

2.97594

PI $(-2.77194, 3.17994)$ He can't eat negative hotdogs.

So $1.485 \pm 2.004(.204)$

0.408816

PI $(1.076184, 1.893816)$

Part III - Application

Questions 20-24 concern the following dataset:

In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
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chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```
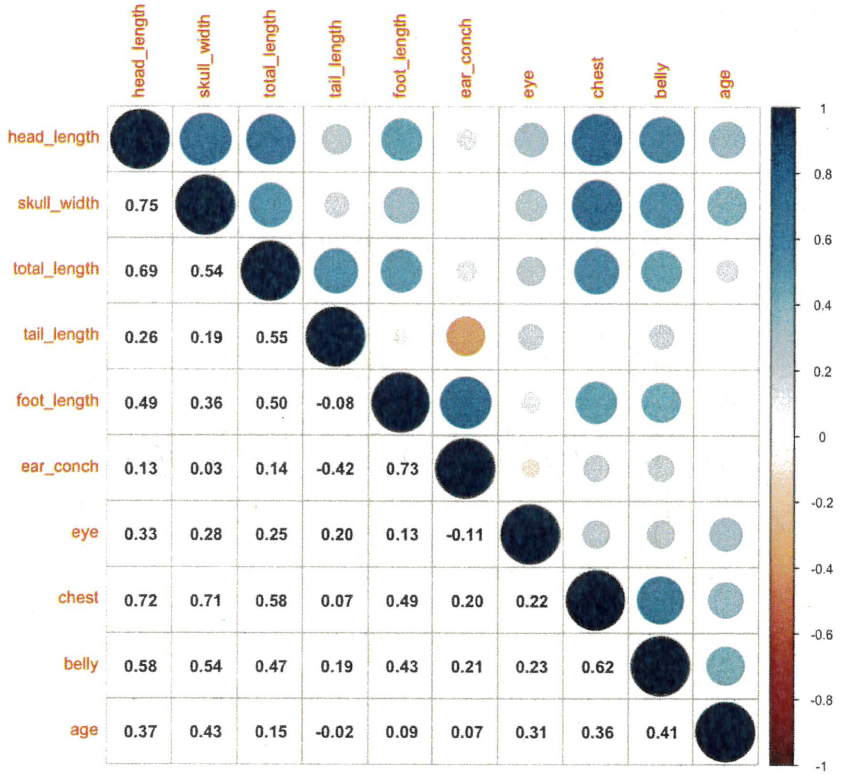

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$.16559 \pm (2.01)(.10599)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

Accounting for all of variables we did not find convincing evidence that belly width is a reliable measurment to determine age since the confidence interval contains 0.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

I believe its because each vary in mean size and each have little significance but with the F-test standardization we are able see that all predictors together help us determine the age of the possum.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

AIC is best when prediction is the main goal and we want lots of predictors. It peralises weak predictors much less than BIC. It is good for a model like this with lots of weak predictors because overall we can still see they are significant all together.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C. Average RMSE across all test-training splits and report

C. Average the predictions and record their value

D. Choose the model with the largest RMSE

C. Choose the model with the smallest RMSE

C. Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

A. Compute each model's AIC and BIC

B. Fit each model using the testing data

B. Fit each model using the training data

B. Initialize vectors for each model that you will use to store their RMSEs

D. Refit each model using the full data set and compare the coefficient estimates to the true values

B. Split the data into test and training sets

B. Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

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Roberts, J Viken

868954587 jviking 949-259-8711

Test: 641159 Form: A Wadsworth, Spencer Gordon

Entry: 06:01pm 10/21/2026 Checked out:

Employee: Jemma C.

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

NS

40142034

053

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Lasso shrinks some coefficients to zero, making a simpler model, while ridge brings coefficients as close as possible to zero, while maintaining all predictors in a given model. Coefficients are shrunk this way

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right). \qquad \left(\frac{.25}{1-.25}\right) \frac{2.4^2}{4} \qquad (1.44)$$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$(X^TX)^{-1}X^TY$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

β comes from the theoretical regression, or the "True population Values" while $\hat{\beta}$ comes from our Data that we use.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

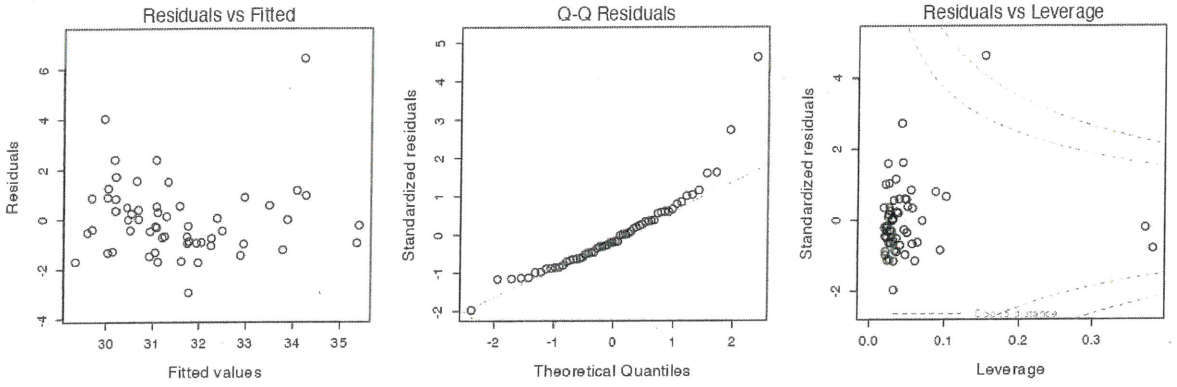
Least Squares is used to estimate regression coefficients by taking the square of every point, penalizing points that are furthest away, bringing everything closer in, and allowing for more test to be accurate.

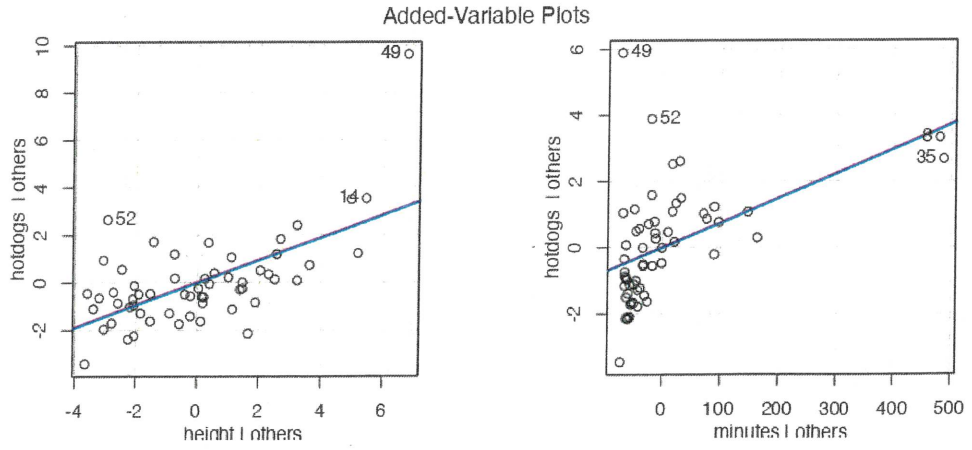
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	Pr($>\|t\|$)
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

.472892

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

Holding all else constant, a one unit increase of Height results in an increase on average of .472892 Hotdogs Eaten.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

It seems that there is a downward trend, however, it is pretty random how the points are dispersed around 0, so I would say it is linear but caution that there is a slight downward trend.

(B) There is independence between each observation

58 competative eaters were chosen [illegible], but most came from a specific places, and it does not specify if they were randomly chosen, putting independence at risk. I would need more information to make a definitive call.

(C) The errors are normally distributed

There is an incredibly small p-value for the Shapiro wilk test, leading us to believe there is evedence that errors are not normally distributed. p value of 2.43e-8

(D) The errors have constant variance (homoscedasticity)

The Breush-Pagan test gives us a value of .071, which is larger than our alpha of .05, so we can assume the errors do have constant varience.

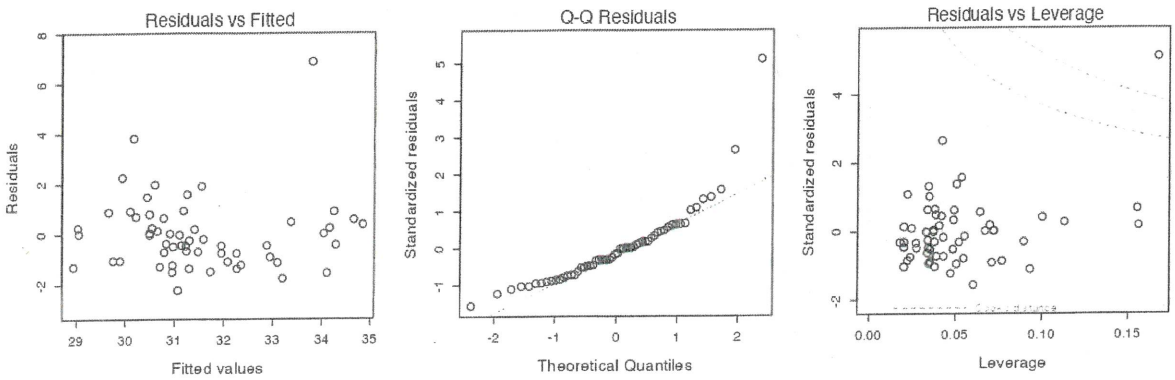
(E) There are no unduly influential points

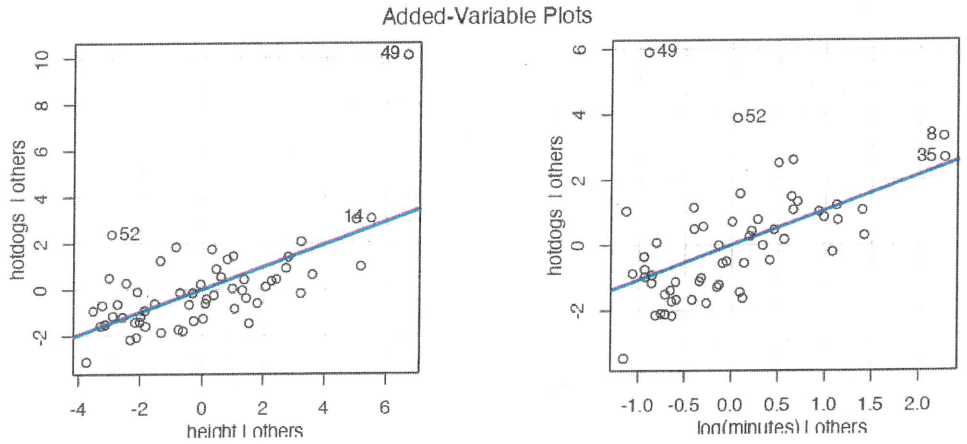
There is a point in the range of Cooks distance on the Residuals vs leverage graph. It would be a good idea to check on it, and make note of it, and make a model without it if necisary. We should not throw the data out because of it though

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

There are many more variables to see before we rule any thing out. A change in a Breush Pagan test like that, where the new number is .28, is significant, but it is not drastic. .028 is less than .05, so we may see heteroscedasticity, but it is also close enough to garner more investigation. We shouldn't just throw it out.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

The CV RMSE is lower, which means our accuracy improved, however, our breush pagan did decrease, showing evedence that another linear regression model assumption was violated.

Question 16: State the estimated value of β_2 and provide an interpretation in context

β_2 log(minutes) = 1.06702 $\quad \frac{1.06702}{100} = .01067702$

for every 1 unit increase in β_2 (minutes practiced), holding all else constant, hotdoge eaten goes .01067702 [illegible] increase.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

HotDogs = -6.05589 + 68(.488334) + log(79)(1.06702)

33.206712 + 11.661838

-6.05589 + 37.868552230

HotDogs = 31.812632230

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

n - p - 1

58 - 2 - 1

55

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

31.813 ± (2.004)(1.485)

31.813 ± 2.97594

95% Prediction interval

(28.83706, 34.789)

Hotdogs eaten

Part III - Application

Questions 20-24 concern the following dataset:

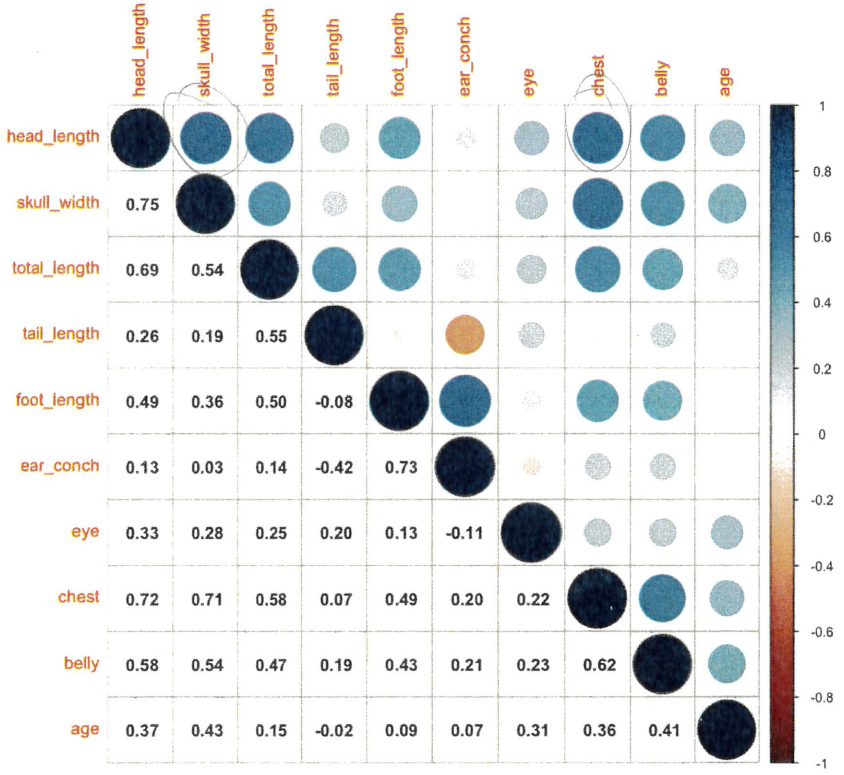
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$.16559 \pm (2.01)(.10599)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

Coefficient for belly width

We are 95% Confident that the population mean lies within

-.023 and .443.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

The number of predictors could be affecting the f-test model significance. More predictors reduce the degrees of freedom, which affect the f test value

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

AIC balances fit and Complexity, so more complexity, often times more predictors are kept. It chooses predictor fits with lower RMSE.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

D Average RMSE across all test-training splits and report

D Average the predictions and record their value

C Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

B Compute each model's AIC and BIC

B Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

B Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM